

Module 01

DevOps Introduction

DevOps end to end services & solutions

Agenda

- > DevOps History
- > Dev vs Ops (The Problem)
- > Waterfall Model
- > Agile Model
- > DevOps Official Birth
- > DevOps Concepts
- > The Future of DevOps

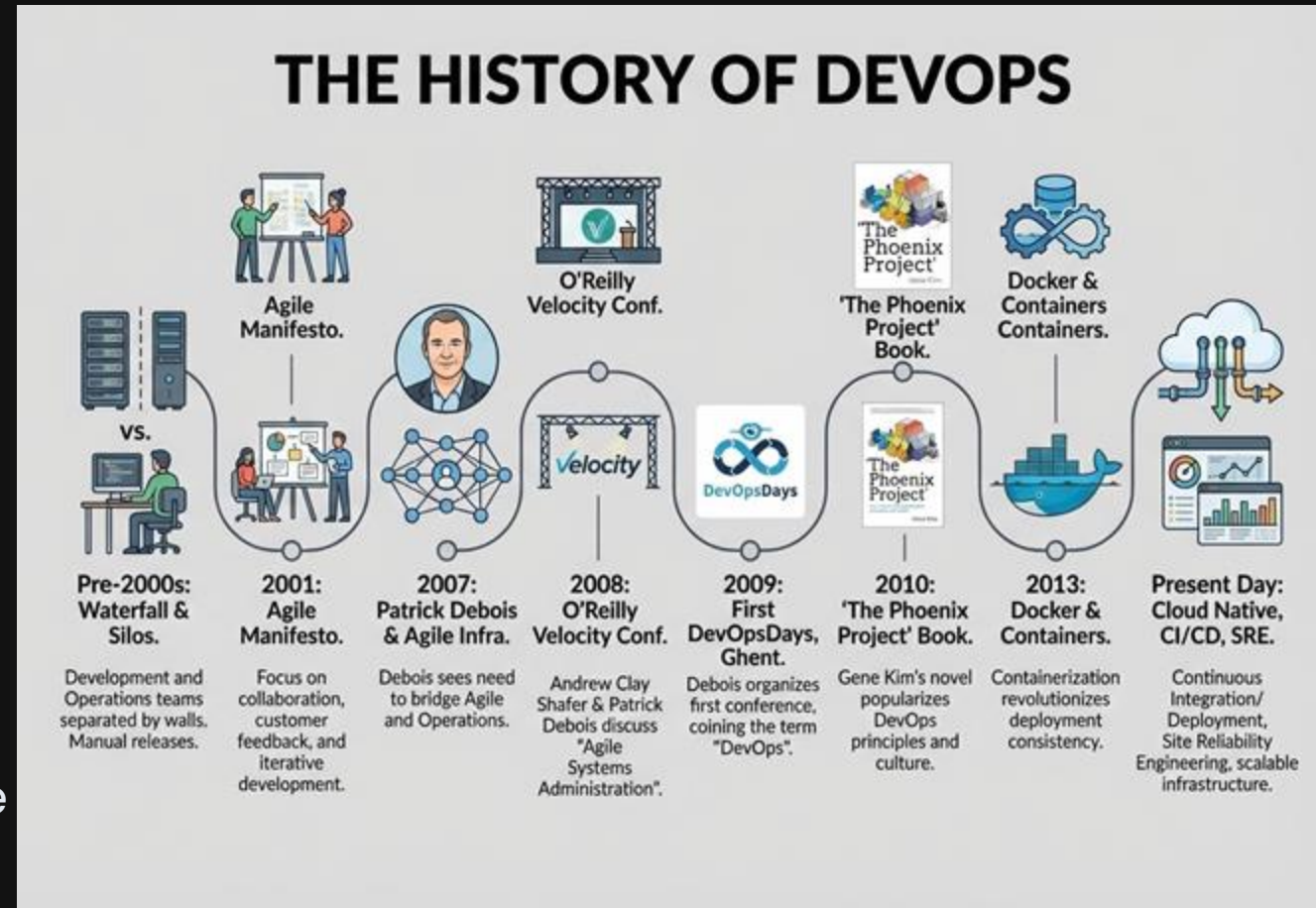
DevOps History

Evolution

The history of DevOps is closely tied to the evolution of software development and IT operations.

- > It represents a shift in how organizations develop, deliver, and maintain software.
- > Aims to bridge the gap between development and operations teams.
- > **Culture:** DevOps is as much about culture as it is about the toolchain.

Software Development Life Cycle(SDLC)



DevOps Official Birth

2009

DevOpsDays Ghent: Patrick Debois organized the first event, coining the term "DevOps".

2010s

The Rise: Companies like Netflix and Amazon demonstrated the power of deploying thousands of times a day.

Goal

To unify software development (Dev) and software operation (Ops).

Dev vs Ops

The Conflict

Ideally, they should work together, but often they are stuck in a "Wall of Confusion".

- > **Dev:** Focuses on change, new features, and speed.
- > **Ops:** Focuses on stability, reliability, and uptime.



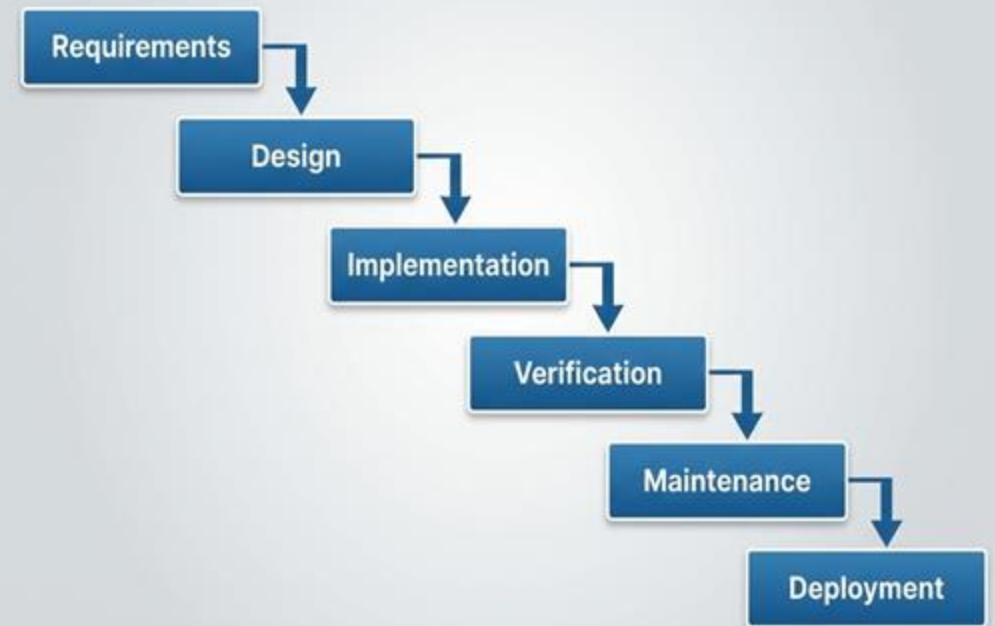
Waterfall Model

↓ Linear Process

Sequential design process where progress flows steadily downwards.

- Requirements -> Design -> Implementation -> Verification -> Maintenance.
- **Drawback:** Inflexible to changes once a phase is complete.

WATERFALL MODEL

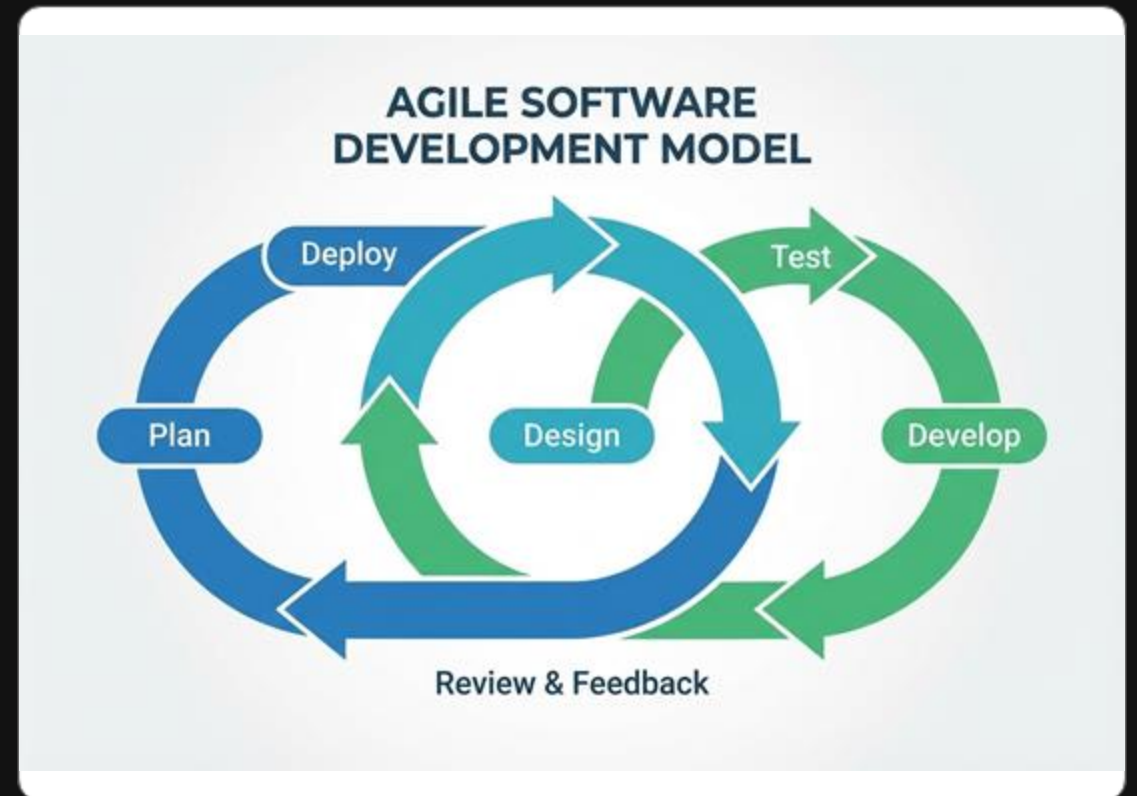


Agile Model

🔄 Iterative Approach

Focuses on iterative development, customer feedback, and small releases.

- > Breaks projects into small "sprints".
- > Encourages collaboration and flexibility.
- > **Limitation:** Often improved Dev speed but Ops was still a bottleneck.



Core DevOps Concepts



CI/CD

Continuous Integration & Continuous Deployment to automate delivery.



IaC

Infrastructure as Code allows managing servers like software.

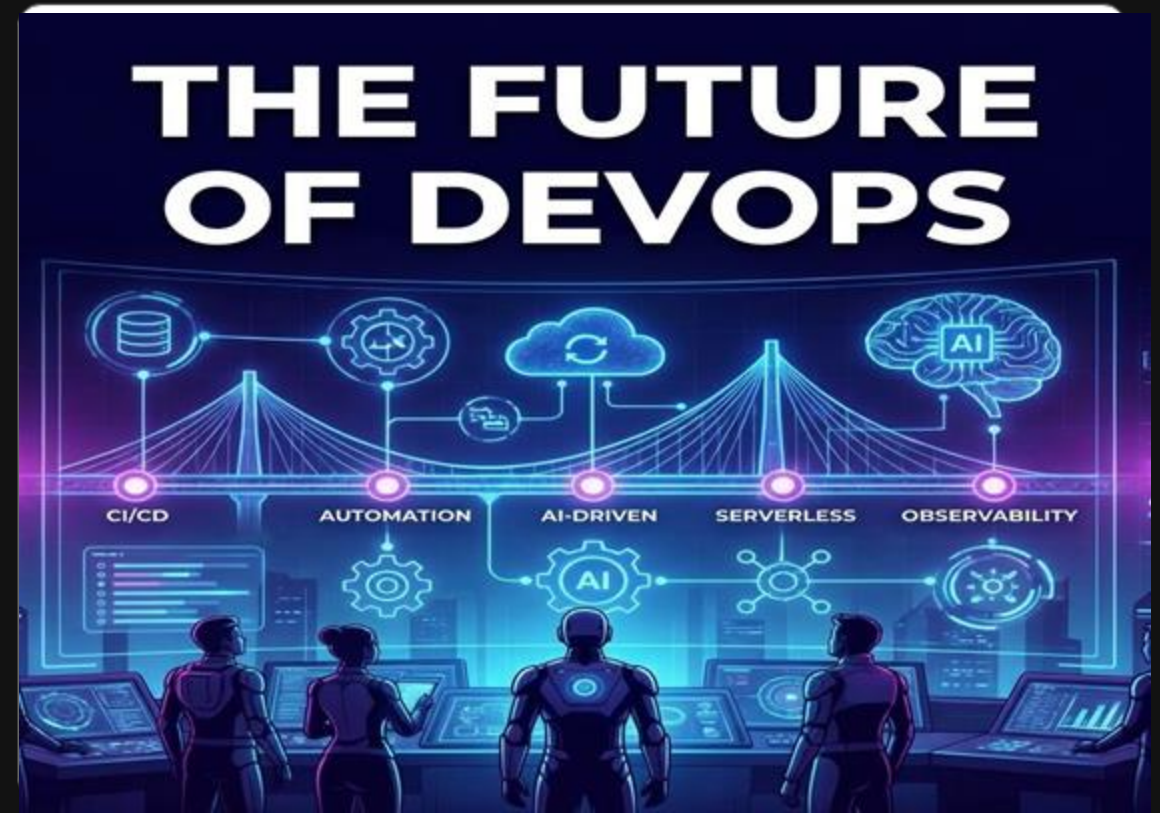


Collaboration

Breaking down silos and fostering a culture of shared responsibility.

The Future of DevOps

- > **AI/ML in DevOps:** Using AI to automate and optimize pipelines (AIOps).
- > **DevSecOps:** Security integrated from the start.
- > **Serverless & Microservices:** Architectures offering greater scalability.
- > **GitOps:** Using Git as the single source of truth for infrastructure.



Questions?

Thank you for participating.

Feel free to reach out:
<https://dinghy.co.il>